Project Title	**Telangana PDS Analytics: Multi-Dimensional Shop Performance Clustering and Anomaly Profiling**
Skills take away From This Project	**Data Integration, Time-Series Analysis, Advanced Feature Engineering, Principal Component Analysis (PCA), K-Means Clustering, DBSCAN, Streamlit Dashboard Development, and Geospatial Data Visualization.**
Domain	**Public Distribution System (PDS) & Supply Chain Logistics**

GUVI HCL Skill Up. Level Up

Problem Statement:

The Telangana State Civil Supplies Department manages thousands of Fair Price Shops (FPS) to ensure food security. Identifying which shops are performing optimally versus those with unusual patterns (e.g., ghost beneficiaries or stock diversion) is challenging at scale. Learners must integrate 7.5 years of transactional, card status, and geospatial data to perform exploratory analysis, segment shops into meaningful performance clusters, and build an interactive dashboard to visualize these insights.

Business Use Cases:

1. **Policy Impact Analysis:** Identifying how the "One Nation One Ration Card" policy changed distribution clusters across urban and rural zones.
2. **Fraud Prevention:** Flagging shops with transaction-to-card ratios that deviate significantly from their peer clusters.
3. **Logistics Optimization:** Identifying "Portability Hubs" that require higher stock replenishment frequency due to high non-local traffic.

Approach:

1. **Data Acquisition & Consolidation:**

- Manually download year-wise CSV files (2023-2025) from the official portal links provided.
- Use a Python loop to combine yearly files into master DataFrames. Perform a Triple-Join on shopNo and distCode to create a unified dataset.

2. **Detailed Exploratory Data Analysis (EDA):**
- Trend Analysis: Plot the growth of "Other Shop Transactions" (portability) from 2023 to 2025.
- Correlation Study: Analyze the relationship between totalRcs (Entitlement) and noOfTrans (Actual Execution).
- Distribution Checks: Use histograms and boxplots to identify the distribution of commodities like Rice and Wheat across districts.
- Seasonality: Check for spikes in transaction volume during specific months (festivals/harvest).

3. **Feature Engineering:**
- Create a Utilization Ratio (Transactions/Total Ration Cards).
- Derive Commodity Intensity (e.g., Rice-to-Wheat ratio).
- Calculate Volatility Features (Standard deviation of transactions over years per shop).
- And more…

4. **Clustering & Analysis:**
- PCA: Reduce high-dimensional data into 2D/3D components for visualization.
- K-Means: Group shops into 4–5 "Behavioral Personas."
- DBSCAN: Specifically identify "Noise" points representing shops with behavior unlike any other cluster.
- Cluster Profiling: For each cluster, calculate the mean of all features (e.g., Cluster 0 = "Urban Hubs", Cluster 1 = "Low-Volume Rural").

5. **Streamlit Dashboard Development:**
- Build an interactive web application allowing users to filter data by District or Year.
- Include a Geospatial Map where shops are color-coded by their Cluster ID.
- Add a Search Tool to input a shopNo and see its performance compared to its cluster average.

Results

- An Interactive Streamlit Dashboard for real-time shop performance monitoring.
- A Cluster Profile Report defining the characteristics of "Typical" vs. "Outlier" shops.
- A Geospatial Hotspot Map highlighting areas with high concentrations of specific cluster types.

Project Evaluation Metrics

- Silhouette Score & Elbow Curve: To validate the optimal number of clusters.
- Cluster Purity: How well the clusters align with known district types (Urban vs. Rural).
- Dashboard Functionality: Responsiveness of filters and clarity of visual insights.

Technical Tags

Python, Pandas, Scikit-Learn, Streamlit, Folium, PCA, K-Means, DBSCAN, Data Wrangling, EDA.

Dataset:

Source: Telangana Open Data Portal
Download CSV files manually from:

- Transactions Data: Link
- Card Status Data: Link
- FPS Locations: Link

Data Set Explanation

- Transactions: Monthly volumes of commodities and portability metrics.
- Card Status: Entitlement context (Who is eligible to receive what).
- Locations: Physical coordinates and status (Active/Inactive).
- More information about data can be found on provided links.

Project Deliverables

1. Jupyter Notebook: Comprehensive EDA and Model Development.
2. Streamlit App Script: (app.py) for the interactive dashboard
3. Final Report: A summary of cluster insights and data-driven recommendations.
4. GitHub repo: Clean folder structure and required files

Project Guidelines

- Visualization Best Practices: In Streamlit, use st.cache_data to ensure the dashboard loads large datasets efficiently.
- Scaling: Always apply StandardScaler before any distance-based modeling (K-Means/DBSCAN).
- Feature Selection: Ensure learners explain why they chose specific features for the final clustering model during their presentation.

** ***In github, upload all your code files.***

Reference:

If you don't know how to approach the project, kindly refer to the project orientation recording provided in this table. (Available in English & Hindi)

Streamlit Doc	https://docs.streamlit.io/library/api-reference
SAMPLE PROJECT FOR REFERENCE(ENGLISH)	Project Excellence Series: Guided…
Streamlit recording (English)	Special session for STREAMLIT(11/08/2…
Project Live Evaluation Metrics	Project Live Evaluation
Capstone Explanation Guideline	Capstone Explanation Guideline
GitHub Reference	GitHub Project Submission Guide
Project Orientation (English)	Telangana_govt_Ration_distribution_clu…
Project Orientation (Hindi)	Project Orientation Session : Telangana…

Timeline: The project must be completed and submitted within **10 days from the assigned date.**

PROJECT DOUBT CLARIFICATION SESSION (PROJECT AND CLASS DOUBTS)

About Session: The Project Doubt Clarification Session is a helpful resource for resolving questions and concerns about projects and class topics. It provides support in understanding project requirements, addressing code issues, and clarifying class concepts. The session aims to enhance comprehension and provide guidance to overcome challenges effectively.
Note: Book the slot at least before 12:00 Pm on the same day

Timing: Monday to Saturday (3:30 PM to 4:30 PM)

Session link and guidelines: Project Doubt Session [DS/AIML]

LIVE EVALUATION SESSION (CAPSTONE AND FINAL PROJECT)

About Session: The Live Evaluation Session for Capstone and Final Projects allows participants to showcase their projects and receive real-time feedback for improvement. It assesses project quality and provides an opportunity for discussion and evaluation.
Note: This form will Open on Saturday and Sunday Only on Every Week

Timing: Monday-Saturday (5:30 PM to 6:30 PM)

Booking link : https://forms.gle/1m2Gsro41fLtZurRA

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